

Efficacy of Mindfulness-Based Stress Reduction in Adults with Chronic Pain: A Randomised Controlled Trial

Authors

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Abstract

Background: Chronic pain affects approximately 20% of adults. Mindfulness-based stress reduction (MBSR) has shown promise in clinical settings. Methods: We conducted a double-blind RCT with $n=120$ participants randomised to MBSR ($n=60$) or waitlist control ($n=60$). Primary outcome was pain intensity (VAS, 0–100) at 8 weeks. Results: MBSR reduced pain intensity ($M=42.3$, $SD=12.1$) compared to control ($M=58.7$, $SD=14.3$); $t(118) = 6.23$, $p < .001$. Cohen's $d = 1.24$. Conclusions: MBSR is an effective adjunct therapy for chronic pain management. Trial registration: ClinicalTrials.gov NCT04521234.

Introduction

Chronic pain is a major public health concern. Previous systematic reviews (DOI:10.1016/j.pain.2020.01.001) indicate heterogeneous outcomes for pharmacological interventions. Non-pharmacological approaches offer potential benefits with fewer side effects.

Methods

Participants were recruited from primary care clinics between January and June 2023. Inclusion criteria: age 18–65, chronic pain > 6 months. Randomisation: computer-generated allocation, concealed envelopes. Both assessors and participants were blinded to allocation. Intervention: 8-week MBSR programme, 2 hours/week. Statistics: two-sample t -tests; significance threshold $\alpha = .05$.

Results

120 participants completed the trial (60 per arm; no dropouts). MBSR group: $M=42.3$, $SD=12.1$, $n=60$. Control group: $M=58.7$, $SD=14.3$, $n=60$. $t(118) = 6.23$, $p < .001$. Effect size Cohen's $d = 1.24$ (large). Secondary: anxiety (GAD-7): $t(118) = 3.14$, $p = .002$.

Competing interests

The authors declare no competing interests.

Data availability

Data are available on the Open Science Framework at osf.io/example123.

References

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